8(a)(i)

$$T = 24 \text{ hours} \quad [1]$$

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{(24\times 60\times 60)}$$

$$= 7.3\times 10^{-5} \text{ rad s}^{-1} \quad [1]$$

(ii)

Gravitational force provides centripetal force

$$\frac{GMm}{r^2} = mr\omega^2 \quad [1]$$

$$r = \sqrt[3]{\frac{GM}{\omega^2}} = \sqrt[3]{\frac{(6.67\times 10^{-11})(5.9\times 10^{24})}{(7.3\times 10^{-5})^2}} \quad [1]$$

$$= 4.2\times 10^{7}\,\text{m} \quad [1]$$

(iii) Communication, weather forecasting or navigation (GPS)

(iv)

Application	Advantage	Disadvantage
Communication Weather Navigation	No break in the signal transmissions as it is fixed position in sky.	High altitude so there is a significant lag time in the signal transmissions.

(b)(i) Gravitational field strength is equal to the negative gravitational potential gradient i.e. $g = -\, d\phi / dr$

(ii) Potential gradient at surface of star S_1 is steeper than that of S_2. [1]
Using relationship in (b)(i), gravitational field strength at the surface of star S_1 is greater than that of star S_2. [1]

(iii) From the graph, when the particle travels from S_2 to S_1, it loses gravitational pe (since it experiences a drop in gravitational potential). [1]

As the total energy of the particle is constant, it gains ke. So its ke at the surface of S_1 is larger than E_k. [1]

(iv) It is the point in which the resultant gravitational field strength is zero. [1]

(v)

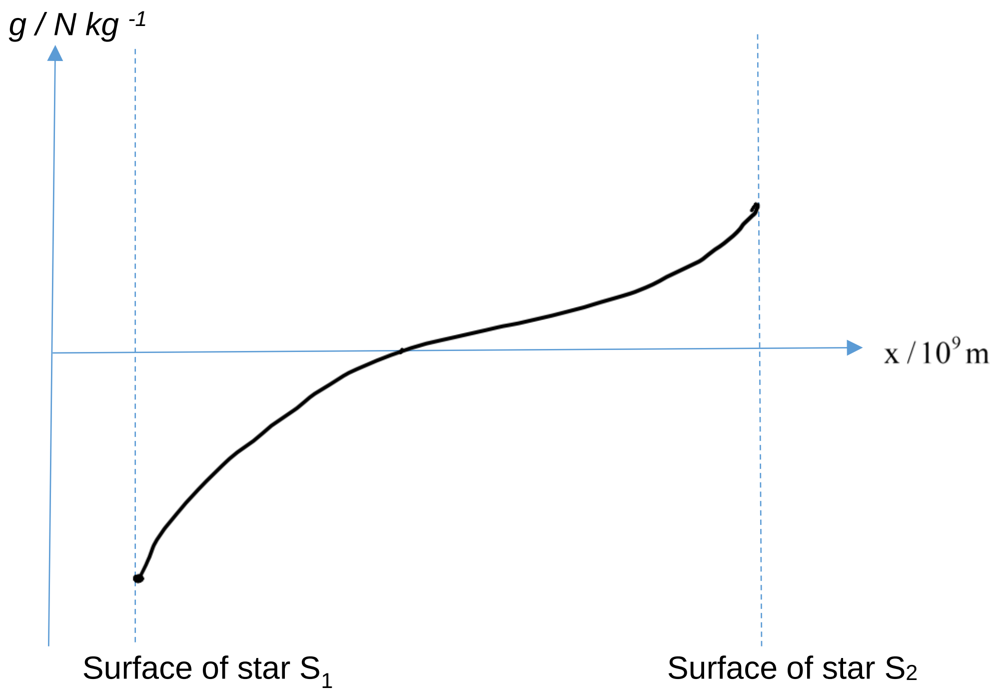
$$\frac{GM_1}{x^2} = \frac{GM_2}{(1.2\times10^{10} - x)^2} \quad [1]$$

From Fig. 8.2, x = 4.8 x 10^9 m [1]

$$\frac{M_1}{M_2} = [\frac{4.8\times10^9}{(1.2\times10^{10} - 4.8\times10^9)}]^2$$

$$= 0.44 \quad [1]$$

(vi)



Correct shape of curve with gravitational field strength at surface of star S_1 greater than S_2. [1]
Field strength is zero at the point of maximum potential. [1]

9(c)(i) Resistance = ρL [1]